Mahbod Afarin

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Professional Summary

Compiler and computer architecture postdoctoral researcher with 5+ years of experience in LLVM/MLIR, hardware acceleration, and hardware/software co-design. Ph.D. in Computer Science with expertise in compiler optimization, hardware accelerators, and graph analytics. Industry research experience at Google developing compiler optimizations for binary code size reduction.

Technical Skills

Programming: C++, C, Python, CUDA, SQL
Compilers: LLVM, MLIR, LLVM-BOLT, LLVM Machine Outliner
Computer Architecture: Hardware/Software Co-Design, Hardware Accelerators, GPU Architecture, PIM
Parallel Computing: OpenMP, CUDA, OpenCL
Hardware Design: Verilog, VHDL, SystemC
EDA & Simulation: Synopsys Design Compiler, Cadence, ModelSim, HSPICE, PSPICE, Quartus, Xilinx ISE

Education

University of California Riverside **Jan. 2020 – Jun. 2025**
Doctor of Philosophy (Ph.D.) in Computer Science *Riverside, California*

- *Thesis: "Redundancy Removal for Accelerating Graph Processing Workloads"*
- *Advisors: Professor Rajiv Gupta & Professor Nael Abu-Ghazaleh*
- *GPA:* **3.86***/4*

Sharif University of Technology **Sep. 2015 – Jan. 2018**
Master of Science (M.Sc.) in Computer Engineering *Tehran, Iran*

- *Thesis: "Improving Life Cycle of SIMT Processors for Approximate Computing" (Thesis Grade: Excellent)*
- *Advisors: Professor Shaahin Hessabi*
- *GPA:* **4***/4 (***19.03***/20) (ranked* **7th** *among 83 computer engineering students)*

Research Experience

University of California San Diego **Jun. 2025 – Present**
Postdoctoral Scholar *San Diego, California*

- Member of the *System Energy Efficiency (SEE) Lab* and the *Processing with Intelligence Storage & Memory (PRISM)* Research Center at UC San Diego, under the supervision of Professor *Tajana Rosing*.
- Developed an MLIR-based compiler framework for dynamic-programming workloads targeting heterogeneous processing-in-memory (PIM) accelerators.
- Designed compiler transformations for tiling, mapping, memory-tier placement, and communication across ReRAM, PCM, and HBM3-based accelerator architectures.

Google **Sep. 2024 – Dec. 2024**
Student Researcher *Sunnyvale, California*

- Member of the *Compiler Optimization* team at Google working on *Inter-procedural Identical Basic Block Folding*.
- Developed scalable inter-procedural identical basic-block folding in LLVM to eliminate redundant code and reduce binary size while preserving runtime performance using profile-guided analysis.

University of California Riverside **Jan. 2020 – Jun. 2025**
Graduate Research Assistant *Riverside, California*

- Member of the *Graph Analytics with Scalability and Performance (GRASP)* and the *RIverside Programming Language & Software Engineering (RIPLE)* centers, supervised by Professor *Rajiv Gupta* & Professor *Nael Abu-Ghazaleh*.
- Developed hardware/software techniques for accelerating evolving and dynamic graph workloads, including specialized graph accelerators and software optimizations.
- Designed graph-processing accelerators and optimization techniques for streaming and evolving graphs, resulting in publications at **MICRO**, **ASPLOS**, **EuroSys**, **BigData** and **IPDPS**.

Sharif University of Technology **Sep. 2015 – Jan. 2018**
Graduate Research Assistant *Tehran, Iran*

- Member of the Very Large Scale Integration Laboratory (VLSI-Lab), supervised by Professor *Shaahin Hessabi*.
- Improved SIMT processor life cycle for approximate computing through techniques that extend hardware longevity.

Selected Publications

C1. [**ASPDAC'27**] **M. Afarin**, T. Lu, T. Rosing, "GenMLIR: An MLIR Compiler for Dynamic Programming Workloads on Tier-Aware 3D-DRAM PIM," *32nd Asia and South Pacific Design Automation Conference.*

C2. [**ASPLOS'27**] X. Routh, A. Noor, A. Kothari, Z. Li, **M. Afarin**, T. Rosing, V. Adve, "ApproxHDC: Approximation Tuning for Hyperdimensional Computing," *International Conference on Architectural Support for Programming Languages and Operating Systems (acceptance rate: **13.68%**).*

C3. [**BigData'26**] B. Gul, **M. Afarin**, F. Ahmed, S. Pinge, T.-H. Lu, T. Rosing, K. Skadron, "HDLouvain: High-Throughput Incremental Clustering for Hyperdimensional Data," *IEEE International Conference on Big Data (acceptance rate: **19.4%**).*

C4. [**BigData'26**] C. Gao, **M. Afarin**, X. Yin, N. Abu-Ghazaleh, R. Gupta, "Sagas: Temporally Consistent Sampling of Evolving Graphs," *IEEE International Conference on Big Data (acceptance rate: **19.4%**).*

C5. [**BigData'26**] C. Gao, X. Yin, **M. Afarin**, N. Abu-Ghazaleh, R. Gupta, "Indexing Evolving Graphs via Query Evolution Prediction," *IEEE International Conference on Big Data (acceptance rate: **19.4%**).*

C6. [**IPDPS'26**] **M. Afarin**, C. Gao, X. Yin, Z. Zhao, N. Abu-Ghazaleh, R. Gupta, "UVVs: Identifying Unchanged Vertex Values in Evolving Graphs via Intersection-Union Analysis," *40th IEEE International Parallel & Distributed Processing Symposium (acceptance rate: **24.5%**).*

C7. [**LCTES'26**] C. Mamatha, **M. Afarin**, R. Gupta, S. Tallam, H. Shen, X. D. Li, "DeduBB: Binary Code Size Reduction via Post-Link Basic Block De-duplication," *Proceedings of the 27th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems.*

C8. [**EuroSys'24**] X. Jiang, **M. Afarin**, Z. Zhao, N. Abu-Ghazaleh, R. Gupta, "Core Graph: Exploiting Edge Centrality to Speedup the Evaluation of Iterative Graph Queries," *Proceedings of the Nineteenth European Conference on Computer Systems (acceptance rate: **15.99%**)* (Contributed Equally with the First Author).

C9. [**MICRO'23**] C. Gao, **M. Afarin**, S. Rahman, N. Abu-Ghazaleh, R. Gupta, "MEGA Evolving Graph Accelerator," *56th Annual IEEE/ACM International Symposium on Microarchitecture (acceptance rate: **22%**)* (Contributed Equally with the First Author).

C10. [**ASPLOS'23**] **M. Afarin**, C. Gao, S. Rahman, N. Abu-Ghazaleh, R. Gupta, "CommonGraph: Graph Analytics on Evolving Data," *International Conference on Architectural Support for Programming Languages and Operating Systems (acceptance rate: **26.66%**).*

C11. [**BigData'23**] A. Mazloumi, **M. Afarin**, R. Gupta, "Expressway: Prioritizing Edges for Distributed Evaluation of Graph Queries," *IEEE International Conference on Big Data (acceptance rate: **17.4%**).*

C12. [**MICRO'21**] S. Rahman, **M. Afarin**, N. Abu-Ghazaleh, R. Gupta, "JetStream: Graph Analytics on Streaming Data with Event-Driven Hardware Accelerator," *54th Annual IEEE/ACM International Symposium on Microarchitecture (acceptance rate: **21.74%**).*

Full publication list: https://mahbod-afarin.github.io/publication/

Awards & Achievements

- Won ***UCR Dissertation Completion Fellowship Award*** at UC Riverside, 2024.
- Received the ***Excellent Service*** badge in all three cycles of ASPLOS'24 Artifact Evaluation, 2024 (Certificate).
- Won ***Dean's Distinguished Fellowship Award*** at University of California, Riverside, 2019.
- ***Ranked 7th*** among 83 Computer Engineering students at Sharif University of Technology (***Top 8%***), 2018.

Professional Services

Audio/Video Chair: ASPLOS'24 Conference
Reviewer: ASPLOS'27, CAL'23, TACO'23, IEEE Transactions on Computers'23, Parallel Computing'23 & 25
Talks: HOPC'23, Society of Women Engineers (UCR, Winter'24), Tulane Uni. (Winter'25), & Binghamton Uni. (Spring'25)
Artifact Evaluation Committee: ASPLOS'25, ASPLOS'24, ISCA'24

Teaching Experience

- Teaching Assistant, ***Compiler Design*** (Summer'21/22/23 and Spring'21/22), UC Riverside, CS Dept., Prof. Rajiv Gupta
- Teaching Assistant, ***System on Chip*** (Spring'18), **Testability** (Fall'17), ***Advanced VLSI*** (Spring'17), ***VLSI*** (Fall'16), Sharif University of Technology, CE Dept., Prof. Shaahin Hessabi.
- Lab Instructor, ***Logic Design Lab*** (Summer 2017), and ***Digital System Design Lab*** (Summer 2016), Sharif University of Technology, Department of Computer Engineering, Prof. Maziar Goudarzi.

References

Professor Tajana Rosing, My Postdoc Supervisor, UC San Diego – (Email | Homepage)
Professor Rajiv Gupta, My Ph.D. Supervisor, UC Riverside – (Email | Homepage)
Professor Nael Abu-Ghazaleh, My Ph.D. Supervisor, UC Riverside – (Email | Homepage)